Development of a 3D-Printable Unibody Compliant Wheel for Enhanced Performance of VIPER in Extreme Polar Environments

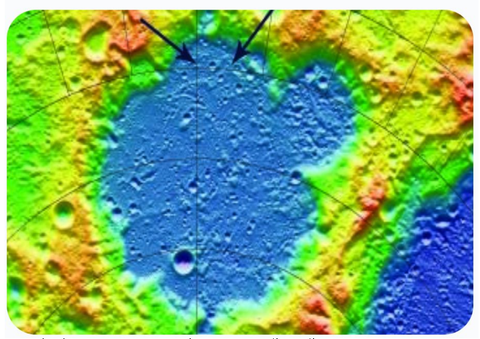
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Abstract

Robust and repairable wheel designs are crucial for lunar rovers in extraterrestrial exploration and settlement. While the existing Volatile Investigating Polar Exploration Rover (VIPER) utilizes rigid, traditionally manufactured wheels, we sought to improve wheel compliance and manufacturing simplicity while maintaining suitability for harsh polar environments. We developed a fully 3D-printable wheel with built-in leaf springs, achieving shock-absorbing properties while maintaining structural simplicity and robustness. This design enables faster travel on rough terrain without extensive suspension systems. Utilizing Finite Element Analysis (FEA) in Fusion 360, we optimized wheel geometry and leaf spring design for maximum compliance without structural failure. The all-metal design addresses the harsh lunar polar environment, ensuring stability across temperatures from 120°C to -249°C and resistance to radiation. Design requirements included minimal displacement under a 174 N rest force, maximum overall displacement, and a safety factor above 3. Titanium emerged as the optimal material among steel and aluminum alternatives, providing the greatest compliance before the risk of failure with a 70.67 mm displacement at a 3.004 safety factor for a 0.5 m diameter wheel. Through iterative design, we developed a unique complex leaf spring geometry leveraging additive manufacturing capabilities. The availability of titanium on the lunar surface as well as the wheel's unibody design could enable on-site 3D printing, supporting self-sufficient settlements without Earth-based material transport. Furthermore, to address variable lighting in permanently shadowed regions, we propose integrating lights with cameras, reducing material usage and space requirements. The extreme temperatures require heating pipes within VIPER's main body for thermal regulation. Our unibody-compliant wheel design enhances adaptability to harsh environments, improves operational capabilities, and supports future lunar settlement initiatives.

Environment

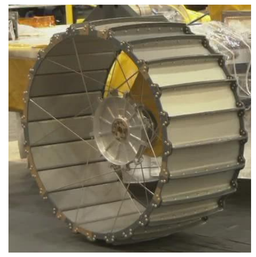
- **Craters**, Lunar Poles
- Sandy with steep incline edges and rocky surfaces.
- **Coldest** measured place in the solar system, **May contain water.**



Wheel Design

0. Original VIPER Wheel

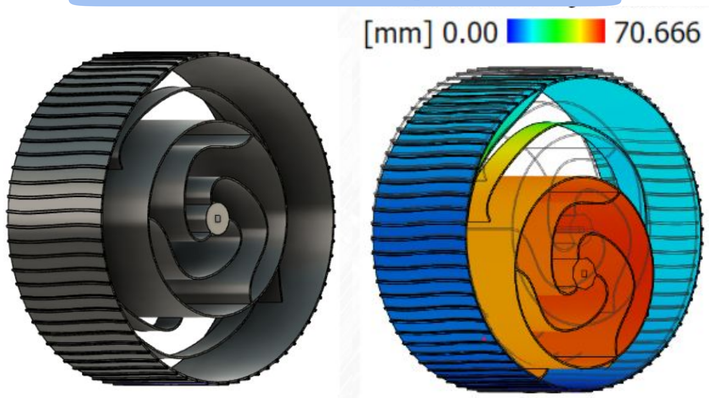
- Diameter of **0.5 meters**
- Simple, lightweight**, inexpensive**, reliable at low speeds
- Rigid aluminum construction with **rigid** spokes
- Shovel like treads are optimized for adaptability for unknown lunar terrain
- **Lacks shock absorption**
- **Cannot be easily 3D printed**



1. Design Criteria

1. Unibody **3D printable compliant** wheel
2. Maximizing Wheel Compression (**Displacement**)
3. Minimizing Wheel deformation at rest
4. Withstand extreme polar environments

2. FEA Analysis



- FEA crucial for rapid design iteration
- Analyzed geometry and materials using Fusion 360
- Key metrics: displacement and safety factor (minimum 3)

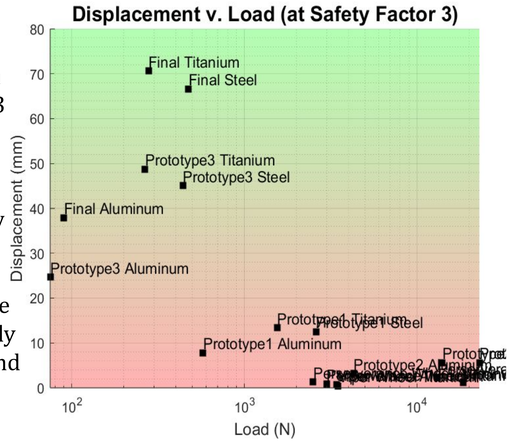
$$\text{Safety Factor} = \frac{\text{Material Strength}}{\text{Actual Stress}} = \frac{\text{Yield Strength}}{\text{Von Mises Equivalent Stress}}$$

3. Data Processing

- Collected simulation data in a spreadsheet and processed in Matlab
- Compared 3 materials: **Aluminum, Steel, Titanium**
- Compared 6 different wheel shapes: Prototypes, Final Design, VIPER and Mars 2020

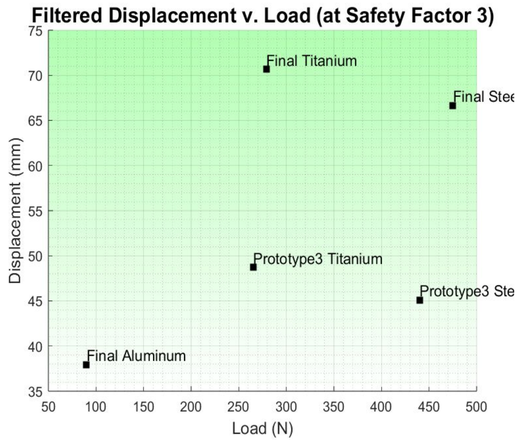
4. Results

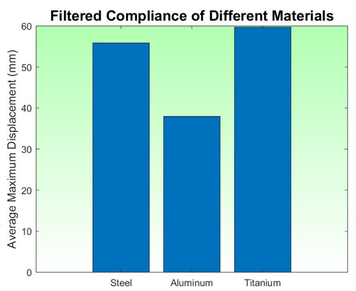
- Plot of Displacement against Load at a Safety Factor of 3
- Clustered data points at the bottom right indicate very low compliance but high strength
- Data points in the red are essentially non compliant and will not absorb impact at high speeds well

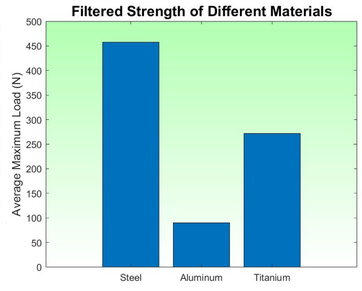


Filter

- Compliance is prioritized over material Strength
- Any data point with less than 5% of compliance (25mm of displacement) was filtered out
- Our Final Design in Titanium performed the best in terms of compliance





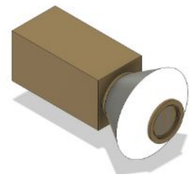


- Titanium is the optimal material for compliance
- Steel is the optimal material for strength
- On Earth, steel will be the cheapest and strongest option
- On the lunar surface Titanium has the most optimal properties
  - Titanium can be harvested on moon for 3D printing

Additional Findings

Camera and Lighting System

- Crater has low-angle to **no illumination**, requiring car-like floodlights
- New design **combines camera and lights** to save space.
- Lights added around the original NavCam for **better illumination**.



Combined lighting and camera CAD design

Thermal Regulation

- Thermal system developed by ACT (Advanced Cooling Technologies)
- Constant conductance heat pipes **spread heat passively**
- Loop **heat pipes** and Lightweight **radiator panels** move heat effectively

Conclusion

Although lunar exploration and development poses many challenges, improvements to the VIPER's **suspension systems, wheels,** and **thermal durability** would improve its effectiveness in studying lunar ice. The proposed designs for a 3D printable compliant wheel and additional improvements to VIPER's heating systems would be instrumental for this ambitious goal. These implementations will significantly bolster the future of space exploration in the Artemis missions and lunar settlement.

Acknowledgements

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